Ali A. Alhejab

Lead AI Engineer · Riyadh, Saudi Arabia · alhejab.ee@gmail.com · github.com/ms802x · Google Scholar

AI engineer focused on generative video and image models, LLMs, and edge AI. I build real-time avatar video models (few-step distillation with DMD and Self Forcing), optimize models for on-device inference on laptop NPUs and mobile, and build the LLM pipelines behind avatar agents with 1.5 s end-to-end latency.

EXPERIENCE

Lead AI Engineer, HUMAIN

May 2025 – Present

Riyadh, Saudi Arabia

- Built the in-house avatar video models, adapting architectures from published research: a half-body generation model distilled to few-step generation with DMD and Self Forcing, running in real time (25 fps, 720p) on a single NVIDIA L4, and a face-inpainting model serving 5+ concurrent streams per L4.
- Optimized a compact avatar model to run on-device on the HUMAIN laptop NPU; deployed on-device models on mobile phones.
- Pretrained and fine-tuned video diffusion models for custom avatar generation; optimized inference for both GPU and CPU deployment.
- Fine-tuned LLMs with LoRA (Gemma E2B, E4B, and 31B) and optimized inference serving to reach 1.5 s end-to-end latency for the conversational avatar.
- Developed real-time AI avatar agents (virtual presenters, meeting facilitators) with STT → LLM → TTS pipelines.
- Built an agent-name-agnostic binary intent classifier (99% accuracy) to route user requests within the conversational pipeline.

AI Engineer, SDAIA

Jun 2024 – May 2025

Saudi Arabia

- Fine-tuned LoRA modules for image and video diffusion models (Flux, Hunyuan, Wan 2.1).
- Fine-tuned LLaMA and T5 for token-by-token image generation/understanding and conversational abilities.
- Optimized inference with TensorRT (vision models), vLLM (LLMs), and TeaCache (diffusion).
- Built multi-GPU / multi-node training pipelines with Hugging Face Accelerate and DeepSpeed.
- Built a sign-language-to-text translation system with T5 and AV-HuBERT (Sign4All Challenge).
- Trained DETR and YOLO detectors for construction-site surveillance (helmets, vehicles, license plates, people).

Solar Energy Systems Researcher, KAUST

Aug 2022 – May 2024

Saudi Arabia

- Developed a single-voltage-sensor fault-localization method for PV systems with the KAUST Solar Center; tested solar-glass technologies with NEOM.

SKILLS

- **Generative models:** Video and image diffusion (Flux, Hunyuan, Wan 2.1); few-step distillation (DMD / DMD2); Self Forcing for autoregressive video; LoRA fine-tuning; inpainting
- **LLMs:** Gemma, LLaMA, T5, LoRA fine-tuning, LLaMA-Factory, vLLM, real-time STT → LLM → TTS agents
- **Efficient inference:** TensorRT, vLLM, TeaCache, torch.compile, FlashAttention, real-time GPU serving (NVIDIA L4), CPU inference
- **Edge AI:** On-device deployment (laptop NPU, mobile); ONNX Runtime, Qualcomm QNN / AI Hub, OpenVINO; quantization, pruning, distillation
- **Training:** PyTorch, Hugging Face, Accelerate, DeepSpeed (multi-GPU / multi-node)
- **Vision:** DETR, YOLO, AV-HuBERT
- **Languages:** Python, C/C++ · Arabic (native), English (advanced)

EDUCATION

M.Sc., Electrical and Computer Engineering Jun 2024

King Abdullah University of Science and Technology (KAUST) · Thuwal, Saudi Arabia

Specialization in Solar Energy Systems · GPA: 3.67/4.0. Thesis: "A Novel Fault Localization Technique for PV Systems Using a Single-Voltage Sensor." Advisor: Prof. Shehab Ahmed.

B.Sc., Electrical Engineering May 2022

King Fahd University of Petroleum and Minerals (KFUPM) · Dhahran, Saudi Arabia

Specialization in Internet of Things · GPA: 3.75/4.0. Research: "Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter." Advisor: Dr. Yaqub Mahnashi.

PUBLICATIONS

1. A. Alhejab et al. "Saudi Sign Language Translation Using T5." *Springer, Cham*, 2026.
2. A. Alhejab, M. Abbasi, S. Ahmed. "A Single-Voltage-Sensor Bypass-Switch-Based Photovoltaic Fault Localization." *IEEE Journal of Photovoltaics*, 2025.
3. O. Khalifa, A. S. Mohammed, A. Alhejab, et al. "Energy-Efficient Aerial Data Aggregation for IoT: From Prototyping to Large-Scale Implementation." *IEEE Transactions on Instrumentation and Measurement*, 2024.
4. A. S. Mohammed, O. Khalifa, A. Alhejab, et al. "Energy-Efficient Aerial Data Aggregation in IoT Networks with WuR." *IEEE International Black Sea Conference on Communications and Networking*, 2023.
5. A. Alhejab, Y. Mahnashi. "A New Synthesis Method for Efficient Reconfigurable Switched-Capacitor Converters." *Authorea Preprints*, 2024.

HONORS

- 1st Place, Abu Dhabi University 11th Research Competition, 2023. Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter.
- 1st Place, KFUPM Senior Design Project EXPO, 2022. Low-Power IoT Data Aggregation System Using a UAV as Data Aggregator.

CERTIFICATIONS

- NVIDIA Certified Associate — Generative AI (Multimodal) & Generative AI (LLM). Dec 2024
- Self-Driving Car Engineer Nanodegree (Misk-sponsored). Jun–Oct 2021

MENTORING

- Mentor, KAUST AI Academy (Deep Learning, Reinforcement Learning), 2023–2024.